

Comprehensive Analysis - Sales Prediction Dataset Analysis



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Objectives of the Analysis

To examine the impact of different advertising channels (TV, Radio, and Newspaper) on product sales using regression analysis.

To determine which medium contributes most significantly to sales and should be prioritized in budget allocation.

To evaluate the overall performance of a multiple linear regression model in predicting sales based on advertising expenditures.

To assess the effect of normalization on model performance and the interpretability of regression coefficients.

To explore how sales predictions change when certain advertising variables (such as TV) are excluded, thereby testing the relative importance of each predictor.

To provide actionable insights and recommendations that can guide the company's advertising strategy for maximizing sales and return on investment.



DATASET OVERVIEW

The dataset consists of **200 advertising campaigns**, each recording expenditure across three media channels (**TV, Radio, and Newspaper**) along with the corresponding sales figures.

- **TV Advertising:** Represents the amount spent on television promotions. This is the largest expenditure category and shows the strongest relationship with sales.
- **Radio Advertising:** Captures spending on radio ads, which has a moderate positive effect on sales.
- **Newspaper Advertising:** Tracks spending on newspaper ads, which contributes very little to sales.
- **Sales:** Indicates the number of product units sold corresponding to the given advertising expenditures.

Data quality checks revealed two missing values in the Radio column, which were imputed using the mean. After cleaning, the dataset had no missing values or duplicates.

Overall, this dataset provides a structured basis for examining how advertising investments across different media channels impact product sales and for building predictive models.

1

Average TV Spend

We analyzed the average amount spent on TV advertising in the dataset.

Result:

Average TV spend = 147.04 units.

Observation:

TV receives the highest investment compared to other channels.

Insight:

Indicates TV is already treated as the primary advertising medium.

Recommendation:

Continue allocating substantial budget to TV advertising.

Conclusion:

TV is the central pillar of the advertising strategy.

Key takeaway:

High average spend highlights the company's reliance on mass-media advertising.

2

Correlation of Radio and Sales

We explored the correlation between radio advertising and sales.

Result:

Correlation = 0.35
(moderate positive).

Observation:

Radio spend moderately boosts sales.

Insight:

Radio helps but impact is smaller than TV.

Recommendation:

Maintain radio as a supporting channel.

Conclusion:

Radio is useful for targeted audiences but not a main driver.

Key takeaway:

Radio works best when combined with TV campaigns for synergy.

3

Highest Impact Advertising Medium

We identified which advertising medium most influences sales.

Result: TV: 4.66, Radio: 1.58, Newspaper: 0.007.

Observation: TV has the strongest impact; Newspaper is negligible.

Insight: TV drives most sales, Radio supports, Newspaper is ineffective.

Recommendation: Increase TV spend, reduce Newspaper, keep Radio as support.

Conclusion: TV should remain the priority for maximum sales impact.

Key takeaway: Newspaper spending offers very low return on investment compared to other channels.

4

Regression Model Fit (All Variables)

We tested multiple
linear regression
with all predictors.

Result:

Predictions closely match
actual sales values.

Observation:

Small deviations exist but
overall accuracy is strong.

Insight:

The regression model is a
reliable tool for sales
prediction.

Recommendation:

Use the model for budget
allocation decisions.

Conclusion:

Multiple regression is
effective in predicting
sales.

Key takeaway:

The model could be
improved further by
adding new advertising or
digital marketing channels.

5

Predicting New Expenditures

We predicted sales for TV=200, Radio=40, Newspaper=50.

Result:

Predicted Sales $\approx$ 19.8 units.

Observation:

Model can forecast sales based on new inputs.

Insight:

Useful for scenario planning with different ad spends.

Recommendation:

Apply the model for forecasting future campaigns.

Conclusion:

Regression helps guide strategic budget planning.

Key takeaway:

This predictive ability supports better resource allocation and risk reduction.

6

Normalization Impact

We examined
model
performance with
normalized data.

Result:

R^2 and errors remain almost unchanged.

Observation:

Coefficients change in scale, not accuracy.

Insight:

Normalization improves interpretability, not performance.

Recommendation:

Normalize when comparing feature importance.

Conclusion:

Use normalization for interpretability, not for accuracy gains.

Key takeaway:

Normalization ensures fair contribution when new features with very different scales are introduced.

7

Radio + Newspaper Only

We tested
regression with
only Radio and
Newspaper.

Result:
Predictions deviate
significantly from actual
sales.

Observation:
Model performs poorly
without TV.

Insight:
Radio contributes
somewhat, Newspaper
almost none.

Recommendation:
Do not rely on Radio +
Newspaper alone for
predictions.

Conclusion:
TV is essential for
accurate sales
forecasting.

Key takeaway:
The exclusion of TV
demonstrates its critical
role as the dominant
driver of sales.

Key Insights

- **TV advertising is the dominant driver of sales**, with the strongest impact compared to Radio and Newspaper.
- **Radio has a moderate positive effect**, useful as a supporting channel but not a main contributor.
- **Newspaper advertising has negligible influence**, offering very low return on investment.
- **Regression model performance is strong** when all three predictors are included, showing reliable predictions.
- **Normalization does not change accuracy**, but it improves interpretability by standardizing coefficients.
- **Excluding TV weakens the model significantly**, proving its essential role in sales forecasting.
- The regression model can be effectively used for **forecasting sales under new advertising budget scenarios**.

Recommendations

Based on the analysis, the following can be recommended :

- **Prioritize TV advertising** in budget allocation to maximize sales impact.
- **Maintain Radio as a supporting channel**, especially for targeted audiences or local markets.
- **Reduce or eliminate Newspaper spend**, reallocating resources to more effective channels.
- **Use regression predictions** to guide advertising budget planning and scenario testing.
- **Apply normalization** when comparing feature importance across predictors or adding new variables.
- **Leverage the model for decision-making**, ensuring strategies are data-driven and ROI-focused.

THANK YOU